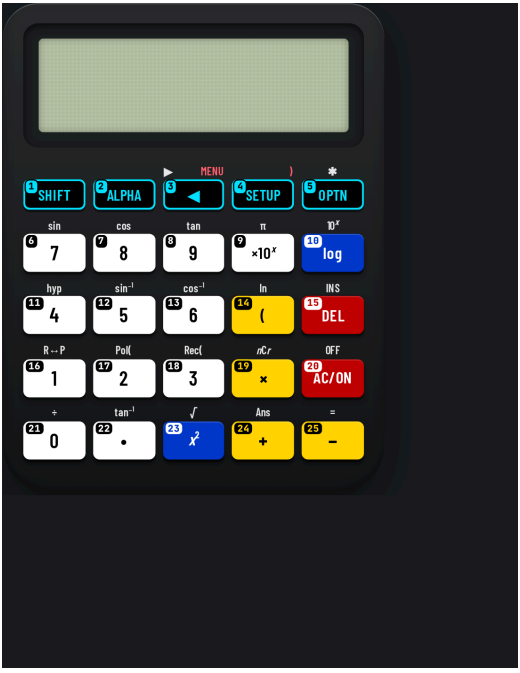
U-45 CALCULATORS

User Manual

Calculator #3 Compact Pocket Scientific (black, blue AC)

Scientific – compact pocket



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#3 — Compact Pocket Scientific (black, blue AC)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic (type the expression as written, then = / EXE).

Chapters

1. Getting started
2. Features
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5. Worked examples
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7. Keyboard shortcuts
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1 · Getting started

Open U-45 CALCULATORS, choose **#3** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#6, #7, #8, #9, #11, #12, #13, #16, #17, #18, #21, #22
op	Operators & execute	#14, #19, #24, #25
fn	Functions	#10, #23
mem	Memory, variables & registers	—
clr	Clear, delete & power	#15, #20
mod	Shift, modes, menus & navigation	#1, #2, #3, #4, #5

2 · Features

44 features: 22 live on the keys, 9 app/UI features, 13 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Compact two-line dot display**
2. LIVE **SHIFT 0 = ÷, SHIFT − = "=", SHIFT + = Ans, SHIFT x^2 = √ (key-saving layout)**
3. LIVE **OPTN menu**
4. LIVE **Pol / Rec conversions**
5. LIVE **Physical keyboard /) Enter fill in the missing keys**
6. LIVE **Square root √**
7. LIVE **Square x^2**
8. LIVE **Common logarithm $\log_{10}$**
9. LIVE **Natural logarithm ln**
10. LIVE **Antilog 10^x**
11. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
12. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
13. LIVE **Hyperbolic sinh / cosh / tanh and their inverses via hyp**
14. LIVE **Rectangular → polar conversion Pol(x,y) → r, θ**
15. LIVE **Polar → rectangular conversion Rec(r,θ) → x, y**
16. LIVE **Combinations nCr**
17. LIVE **π constant to full internal precision**
18. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
19. LIVE **Ans recall of the last result; operators auto-insert Ans**

20. LIVE MODE / SETUP: Fix / Sci / Norm display, angle unit, fraction format
21. LIVE Backspace / DEL editing of the current entry
22. LIVE INS insert mode for editing in the middle of an expression
23. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
24. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
25. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
26. APP/UI Show / hide key numbers with one button (remembered between sessions)
27. APP/UI Copy the current display text to the clipboard
28. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
29. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
30. APP/UI Reference photo viewer and tool-tip name on every key
31. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
32. PLANNED Statistics mode
33. PLANNED Base-N mode
34. PLANNED Equation solver
35. PLANNED Fraction template
36. PLANNED Memory M+ / M–
37. PLANNED Table mode
38. PLANNED Constants

39. PLANNED **Unit conversions**

40. PLANNED **Exam mode (locks memory & shows indicator)**

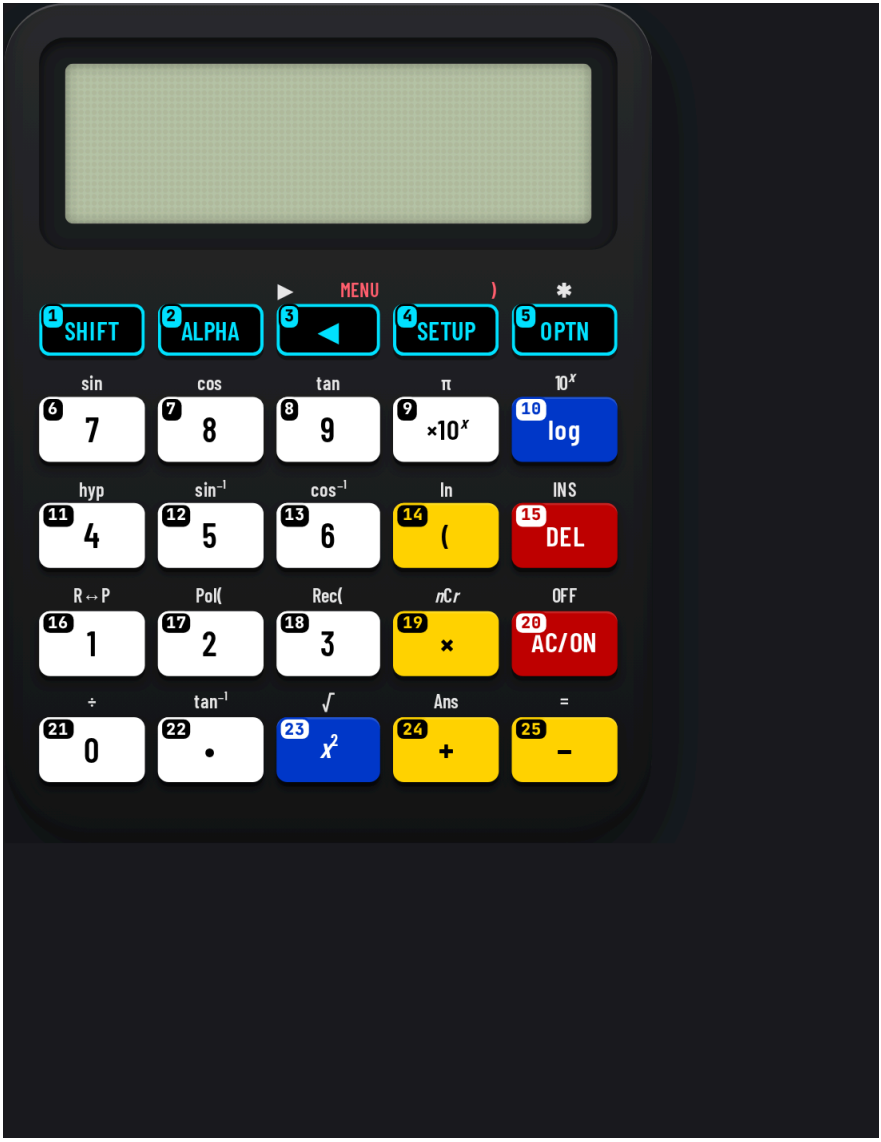
41. PLANNED **Answer history export to text**

42. PLANNED **Speech read-out of results**

43. PLANNED **Colour-blind safe key theme**

44. PLANNED **Custom shortcut keys**

3 · Numbered key map



Calculator #3 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	SHIFT	·	·	·	mod
2	ALPHA	·	·	·	mod
3	◀	▶	MENU	·	mod
4	SETUP	·	)	·	mod
5	OPTN	✱	·	·	mod
6	7	sin	·	·	num
7	8	cos	·	·	num
8	9	tan	·	·	num
9	×10x	π	·	·	num
10	log	10x	·	·	fn
11	4	hyp	·	·	num
12	5	$\sin^{-1}$	·	·	num
13	6	$\cos^{-1}$	·	·	num
14	(	ln	·	·	op
15	DEL	INS	·	·	clr
16	1	R ↔ P	·	·	num
17	2	Pol(	·	·	num
18	3	Rec(	·	·	num
19	×	nCr	·	·	op
20	AC/ON	OFF	·	·	clr
21	0	÷	·	·	num
22	•	$\tan^{-1}$	·	·	num
23	x^2	√	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	+	Ans	·	·	op
25	−	=	·	·	op

4 · Key-by-key reference

Every key on calculator #3, in number order, with every function it carries in this app.

#1 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#2 ALPHA `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#3 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

shifted (2nd / f / SHIFT layer) (press #1 first): ▶ — Cursor ▶ — move right in the line

alpha layer (press #2 first): MENU — MODE — choose calculation mode

Keyboard: `ArrowLeft`

#4 SETUP `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

alpha layer (press #2 first):) — Closing bracket). Closing brackets at the end of a line may be omitted

#5 OPTN `Shift, modes, menus & navigation`

Primary: OPTN — options for the current mode

shifted (2nd / f / SHIFT layer) (press #1 first): ✱ — Multiplication ×

#6 7 `Digits & number entry`

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #1 first): sin — Sine (uses the current angle unit)

Keyboard: 7

#7 8 `Digits & number entry`

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #1 first): cos — Cosine (current angle unit)

Keyboard: 8

#8 9 `Digits & number entry`

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #1 first): tan — Tangent (current angle unit)

Keyboard: 9

#9 ×10x `Digits & number entry`

Primary: ×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23

shifted (2nd / f / SHIFT layer) (press #1 first): π — π (3.14159265…) as an exact constant

Keyboard: e

#10 log `Functions`

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #1 first): 10x — 10^{x} — antilog

Keyboard: l

#11 4 `Digits & number entry`

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #1 first): hyp — Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

Keyboard: 4

#12 5 `Digits & number entry`

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #1 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

Keyboard: 5

#13 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #1 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

Keyboard: 6

#14 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #1 first): ln — Natural logarithm ln

Keyboard: (

#15 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #1 first): INS — INS — insert mode at the cursor

Keyboard: Backspace Delete

#16 1 Digits & number entry

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #1 first): R↔P — Rectangular → polar Pol(x,y)

Keyboard: 1

#17 2 Digits & number entry

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #1 first): Pol(— Rectangular → polar Pol(x,y)

Keyboard: 2

#18 3 Digits & number entry

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #1 first): Rec(— Polar → rectangular Rec(r,θ)

Keyboard: 3

#19 × Operators & execute

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #1 first): nCr — Combinations nCr

Keyboard: * X

#20 AC/ON Clear, delete & power

Primary: ON / AC — power on, or clear all

shifted (2nd / f / SHIFT layer) (press #1 first): OFF — OFF — power off (state is lost; press ON to restart)

#21 0 Digits & number entry

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #1 first): ÷ — Division ÷

Keyboard: 0

#22 • Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #1 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

Keyboard: . ,

#23 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #1 first): √ — Square root √

Keyboard: q

#24 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #1 first): Ans — Ans — recall the last answer into the current expression

Keyboard: +

#25 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #1 first): = — Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: -

5 · Worked examples

17 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#16 1 → #17 2 → #24 + → #18 3 → #1 SHIFT → #5 ✱ → #11 4 → #1 SHIFT → #25 =

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#14 (→ #16 1 → #17 2 → #24 + → #7 8 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #11 4 → #1 SHIFT → #25 =

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate **(3 + 4) × (9 − 2)**. Press **Reset** first.

#14 (→ #18 3 → #24 + → #11 4 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #8 9 → #25 − → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #25 =

Display shows `49` — expected 49. ✔ verified on the app.

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press **Reset** first.

#16 1 → #17 2 → #23 x^2 → #1 SHIFT → #25 =

Display shows `144` — expected 144. ✔ verified on the app.

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate $\sqrt{144}$. Press **Reset** first.

#1 SHIFT → #23 √ → #16 1 → #11 4 → #11 4 → #1 SHIFT → #25 =

Display shows `12` — expected 12. ✔ verified on the app.

The square-root key returns the principal (positive) root.

Example 5.6 — Trigonometry (degrees)

Calculate **tan 45° (Deg mode)**. Press **Reset** first.

#1 SHIFT → #8 tan → #11 4 → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #25 =

Display shows `1` — expected 1. ✔ verified on the app.

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.7 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#10 log → #16 1 → #21 0 → #21 0 → #21 0 → #2 ALPHA → #4) → #1 SHIFT → #25 =

Display shows `3` — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.8 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#1 SHIFT → #14 ln → #16 1 → #2 ALPHA → #4) → #1 SHIFT → #25 =

Display shows `0` — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.9 — Antilog 10^x

Calculate 10^2. Press **Reset** first.

#1 SHIFT → #10 10^x → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #25 =

Display shows 100 — expected 100. ✔ verified on the app.

10^x is the inverse of log.

Example 5.10 — Scientific notation

Calculate 6×10^2 × 3. Press **Reset** first.

#13 6 → #9 ×10^x → #17 2 → #1 SHIFT → #5 ✱ → #18 3 → #1 SHIFT → #25 =

Display shows 1800 — expected 1800. ✔ verified on the app.

The exponent key enters ×10^n without typing × 10 ^.

Example 5.11 — Area of a circle πr^2

Calculate $\pi \times 3^2$. Press **Reset** first.

#1 SHIFT → #9 π → #1 SHIFT → #5 ✱ → #18 3 → #23 x^2 → #1 SHIFT → #25 =

Display shows 28.27433388 — expected 28.27433388. ✔ verified on the app.

π is stored to full precision — never type 3.14.

Example 5.12 — Combinations nCr

Calculate 5C2. Press Reset first.

#12 5 → #1 SHIFT → #19 *nCr* → #17 2 → #1 SHIFT → #25 =

Display shows 10 **— expected 10. ✔ verified on the app.**

nCr counts selections where order does not matter.

Example 5.13 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#16 1 → #21 0 → #21 0 → #1 SHIFT → #5 ✱ → #11 4 → #1 SHIFT → #25 = → #1 SHIFT → #21 ÷ → #17 2 → #1 SHIFT → #25 =

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.14 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#14 (→ #11 4 → #24 + → #7 8 → #24 + → #16 1 → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #18 3 → #1 SHIFT → #25 =

Display shows 9 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.15 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#12 5 → #21 0 → #25 − → #11 4 → #1 SHIFT → #5 ✱ → #13 6 → #24 + → #6 7 → #1 SHIFT → #25 =

Display shows 33 **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.16 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#18 3 → #13 6 → #21 0 → #1 SHIFT → #21 ÷ → #11 4 → #1 SHIFT → #21 ÷ → #8 9 → #1 SHIFT → #25 =

Display shows 10 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.17 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#14 (→ #14 (→ #18 3 → #24 + → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #11 4 → #25 − → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #16 1 → #21 0 → #1 SHIFT → #25 =

Display shows 3 **— expected 3. ✔ verified on the app.**

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

24 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #3 → ▶; #2 then #3 → MENU; #2 then #4 →); #1 then #5 → ✱.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #25.

Hack 13: Type faster on a computer

20 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
ArrowLeft	#3	◀	Cursor ◀ — move left in the line
7	#6	7	Digit 7
8	#7	8	Digit 8
9	#8	9	Digit 9
e	#9	×10x	×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23
l	#10	log	Common logarithm $\log_{10}$
4	#11	4	Digit 4
5	#12	5	Digit 5
6	#13	6	Digit 6
(	#14	(	Opening bracket (
Backspace Delete	#15	DEL	DEL — delete the character at the cursor
1	#16	1	Digit 1
2	#17	2	Digit 2
3	#18	3	Digit 3
* x	#19	×	Multiplication ×
0	#21	0	Digit 0
. ,	#22	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
q	#23	x^2	Square x^2
+	#24	+	Addition +
-	#25	–	Subtraction –

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (91)

Work each one on calculator #3, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (1 + 2) ÷ 4 = _____
14. (250 + 350) ÷ 12 = _____
15. (5 + 6) × (10 − 3) = _____
16. (12 + 3) × (8 − 5) = _____

17. $(1.5 + 2.5) \times (7 - 4) =$ _____

18. $(20 + 30) \times (15 - 5) =$ _____

19. $(9 + 1) \times (100 - 1) =$ _____

20. $(11 + 13) \times (17 - 9) =$ _____

21. $25^2 =$ _____

22. $7^2 =$ _____

23. $1.5^2 =$ _____

24. $19^2 =$ _____

25. $33^2 =$ _____

26. $0.5^2 =$ _____

27. $125^2 =$ _____

28. $64^2 =$ _____

29. $\sqrt{625} =$ _____

30. $\sqrt{2} =$ _____

31. $\sqrt{50} =$ _____

32. $\sqrt{1024} =$ _____

33. $\sqrt{3} =$ _____

34. $\sqrt{9801} =$ _____

35. $\sin 90°$ **(Deg mode)** = _____

36. $\cos 0°$ **(Deg mode)** = _____

37. log 100 = _____

38. log 10000 = _____

39. ln 10 = _____

40. ln 2 = _____

41. 10^3 = _____

42. 2×10^3 × 4 = _____

43. 1.5×10^4 × 2 = _____

44. 3×10^5 × 3 = _____

45. 4.5×10^2 × 2 = _____

46. $\pi \times 10^2$ = _____

47. $\pi \times 2.5^2$ = _____

48. $\pi \times 7^2$ = _____

49. $\pi \times 1.2^2$ = _____

50. 10C3 = _____

51. 6C6 = _____

52. 15 × 6 = , then Ans ÷ 9 = _____

53. 7.5 × 8 = , then Ans ÷ 3 = _____

54. 250 × 12 = , then Ans ÷ 100 = _____

55. (12 + 15 + 18) ÷ 3 = _____

56. (70 + 85 + 91) ÷ 3 = _____

57. (2.5 + 3.5 + 6) ÷ 3 = _____

58. (100 + 200 + 600) ÷ 3 = _____

59. 100 − 3 × 12 + 9 = _____

60. 18 − 2 × 5 + 40 = _____

61. 7.5 − 1.5 × 3 + 2 = _____

62. 64 − 8 × 4 + 10 = _____

63. 300 − 25 × 4 + 55 = _____

64. 9 − 3 × 3 + 1 = _____

65. 45 − 5 × 5 + 15 = _____

66. 1000 ÷ 8 ÷ 5 = _____

67. 96 ÷ 2 ÷ 3 = _____

68. 7200 ÷ 60 ÷ 60 = _____

69. 500 ÷ 4 ÷ 25 = _____

70. 81 ÷ 3 ÷ 3 = _____

71. ((10 + 2) × 3 − 6) ÷ 6 = _____

72. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

73. ((20 + 30) × 2 − 25) ÷ 3 = _____

74. ((7 + 8) × 9 − 15) ÷ 12 = _____

75. 12 + 3 × 4 = _____

76. (12 + 8) ÷ 4 = _____

77. (3 + 4) × (9 − 2) = _____

78. 12^2 = _____

79. √144 = _____

80. tan 45° (Deg mode) = _____

81. log 1000 = _____

82. ln 1 = _____

83. 10^2 = _____

84. 6×10^2 × 3 = _____

85. π × 3^2 = _____

86. 5C2 = _____

87. 100 × 4 = , then Ans ÷ 2 = _____

88. (4 + 8 + 15) ÷ 3 = _____

89. 50 − 4 × 6 + 7 = _____

90. 360 ÷ 4 ÷ 9 = _____

91. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (39)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “ALPHA” (ALPHA)? _____
3. Which key is “◀” (Cursor ◀)? _____

4. Which key is “SETUP” (SETUP)? _____
5. Which key is “OPTN” (OPTN)? _____
6. Which key is “×10x” (×10^{x} / EXP / EE)? _____
7. Which key is “log” (Common logarithm $\log_{10}$)? _____
8. Which key is “(” (Opening bracket)? _____
9. Which key is “DEL” (DEL)? _____
10. Which key is “×” (Multiplication ×)? _____
11. Which key is “AC/ON” (ON / AC)? _____
12. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____
13. Which key is “x^2” (Square x^2)? _____
14. Which key is “+” (Addition +)? _____
15. Which key is “−” (Subtraction −)? _____
16. How do you reach “▶”? _____
17. How do you reach “MENU”? _____
18. How do you reach “)”? _____
19. How do you reach “✱”? _____
20. How do you reach “sin”? _____
21. How do you reach “cos”? _____
22. How do you reach “tan”? _____
23. How do you reach “π”? _____

24. How do you reach “10x”? _____

25. How do you reach “hyp”? _____

26. How do you reach “$\sin^{-1}$”? _____

27. How do you reach “$\cos^{-1}$”? _____

28. How do you reach “ln”? _____

29. How do you reach “INS”? _____

30. How do you reach “R ↔ P”? _____

31. How do you reach “Pol(”? _____

32. How do you reach “Rec(”? _____

33. How do you reach “nCr”? _____

34. How do you reach “OFF”? _____

35. How do you reach “÷”? _____

36. How do you reach “$\tan^{-1}$”? _____

37. How do you reach “√”? _____

38. How do you reach “Ans”? _____

39. How do you reach “=”? _____

Appendix A — Drill answers

Set A

1. 55 (display `55`)

#17 2 → #12 5 → #24 + → #12 5 → #1 SHIFT → #5 ✱ → #13 6 → #1 SHIFT → #25 =

2. 79 (display `79`)

#6 7 → #24 + → #7 8 → #1 SHIFT → #5 ✱ → #8 9 → #1 SHIFT → #25 =

3. 282 (display `282`)

#16 1 → #12 5 → #21 0 → #24 + → #16 1 → #17 2 → #1 SHIFT → #5 ✱ → #16 1 → #16 1 → #1 SHIFT → #25 =

4. 213 (display `213`)

#18 3 → #24 + → #16 1 → #11 4 → #1 SHIFT → #5 ✱ → #16 1 → #12 5 → #1 SHIFT → #25 =

5. 90 (display `90`)

#11 4 → #7 8 → #24 + → #13 6 → #1 SHIFT → #5 ✱ → #6 7 → #1 SHIFT → #25 =

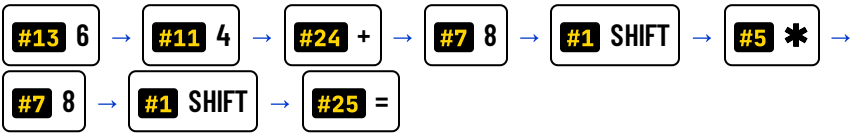
6. 1100 (display `1100`)

#16 1 → #21 0 → #21 0 → #21 0 → #24 + → #17 2 → #12 5 → #1 SHIFT → #5 ✱ → #11 4 → #1 SHIFT → #25 =

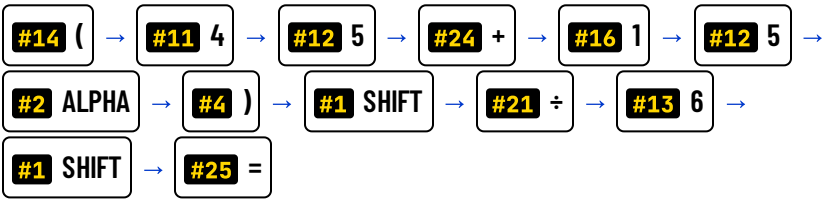
7. 82 (display `82`)



8. 128 (display `128`)



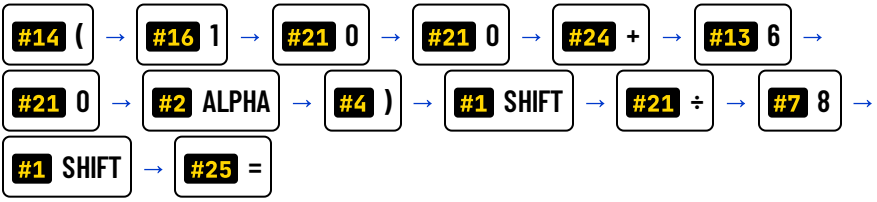
9. 10 (display `10`)



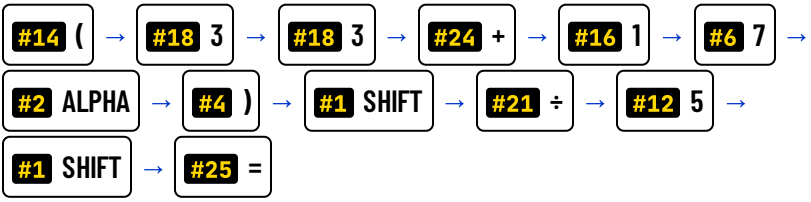
10. 8 (display `8`)



11. 20 (display `20`)



12. 10 (display `10`)



13. 0.75 (display `0.75`)

#14 (→ #16 1 → #24 + → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #11 4 → #1 SHIFT → #25 =

14. 50 (display `50`)

#14 (→ #17 2 → #12 5 → #21 0 → #24 + → #18 3 → #12 5 → #21 0 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #16 1 → #17 2 → #1 SHIFT → #25 =

15. 77 (display `77`)

#14 (→ #12 5 → #24 + → #13 6 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #16 1 → #21 0 → #25 − → #18 3 → #2 ALPHA → #4) → #1 SHIFT → #25 =

16. 45 (display `45`)

#14 (→ #16 1 → #17 2 → #24 + → #18 3 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #7 8 → #25 − → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #25 =

17. 12 (display `12`)

#14 (→ #16 1 → #22 • → #12 5 → #24 + → #17 2 → #22 • → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #6 7 → #25 − → #11 4 → #2 ALPHA → #4) → #1 SHIFT → #25 =

18. 500 (display `500`)

#14 (→ #17 2 → #21 0 → #24 + → #18 3 → #21 0 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #16 1 → #12 5 → #25 − → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #25 =

19. 990 (display `990`)

#14 (→ #8 9 → #24 + → #16 1 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #16 1 → #21 0 → #21 0 → #25 − → #16 1 → #2 ALPHA → #4) → #1 SHIFT → #25 =

20. 192 (display `192`)

#14 (→ #16 1 → #16 1 → #24 + → #16 1 → #18 3 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #16 1 → #6 7 → #25 − → #8 9 → #2 ALPHA → #4) → #1 SHIFT → #25 =

21. 625 (display `625`)

#17 2 → #12 5 → #23 x^2 → #1 SHIFT → #25 =

22. 49 (display `49`)

#6 7 → #23 x^2 → #1 SHIFT → #25 =

23. 2.25 (display `2.25`)

#16 1 → #22 • → #12 5 → #23 x^2 → #1 SHIFT → #25 =

24. 361 (display `361`)

`#16` 1 → `#8` 9 → `#23` x^2 → `#1` SHIFT → `#25` =

25. 1089 (display `1089`)

`#18` 3 → `#18` 3 → `#23` x^2 → `#1` SHIFT → `#25` =

26. 0.25 (display `0.25`)

`#21` 0 → `#22` • → `#12` 5 → `#23` x^2 → `#1` SHIFT → `#25` =

27. 15625 (display `15625`)

`#16` 1 → `#17` 2 → `#12` 5 → `#23` x^2 → `#1` SHIFT → `#25` =

28. 4096 (display `4096`)

`#13` 6 → `#11` 4 → `#23` x^2 → `#1` SHIFT → `#25` =

29. 25 (display `25`)

`#1` SHIFT → `#23` √ → `#13` 6 → `#17` 2 → `#12` 5 → `#1` SHIFT → `#25` =

30. 1.41421356 (display `1.414213562`)

`#1` SHIFT → `#23` √ → `#17` 2 → `#1` SHIFT → `#25` =

31. 7.07106781 (display `7.071067812`)

`#1` SHIFT → `#23` √ → `#12` 5 → `#21` 0 → `#1` SHIFT → `#25` =

32. 32 (display `32`)

`#1` SHIFT → `#23` √ → `#16` 1 → `#21` 0 → `#17` 2 → `#11` 4 → `#1` SHIFT → `#25` =

33. 1.73205081 (display `1.732050808`)

`#1` SHIFT → `#23` √ → `#18` 3 → `#1` SHIFT → `#25` =

34. 99 (display `99`)

`#1` SHIFT → `#23` √ → `#8` 9 → `#7` 8 → `#21` 0 → `#16` 1 → `#1` SHIFT → `#25` =

35. 1 (display `1`)

`#1` SHIFT → `#6` sin → `#8` 9 → `#21` 0 → `#2` ALPHA → `#4`) → `#1` SHIFT → `#25` =

36. 1 (display `1`)

`#1` SHIFT → `#7` cos → `#21` 0 → `#2` ALPHA → `#4`) → `#1` SHIFT → `#25` =

37. 2 (display `2`)

`#10` log → `#16` 1 → `#21` 0 → `#21` 0 → `#2` ALPHA → `#4`) → `#1` SHIFT → `#25` =

38. 4 (display `4`)

`#10` log → `#16` 1 → `#21` 0 → `#21` 0 → `#21` 0 → `#21` 0 → `#2` ALPHA → `#4`) → `#1` SHIFT → `#25` =

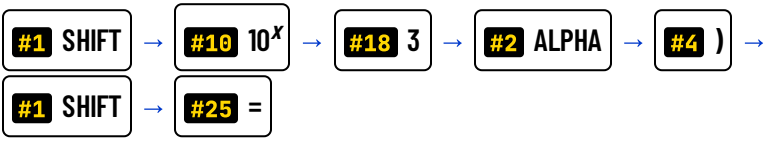
39. 2.30258509 (display `2.302585093`)

`#1` SHIFT → `#14` ln → `#16` 1 → `#21` 0 → `#2` ALPHA → `#4`) → `#1` SHIFT → `#25` =

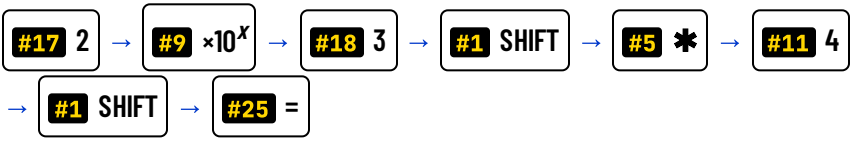
40. 0.69314718 (display `0.6931471806`)



41. 1000 (display `1000`)



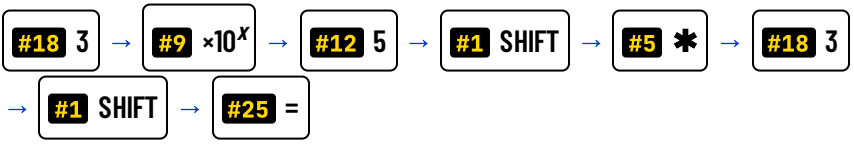
42. 8000 (display `8000`)



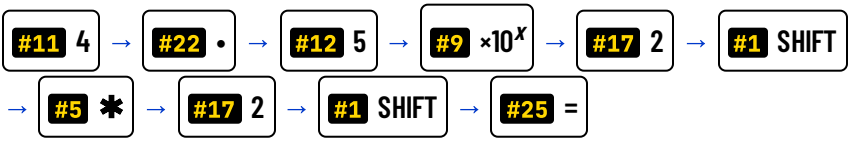
43. 30000 (display `30000`)



44. 900000 (display `900000`)



45. 900 (display `900`)



46. 314.15926536 (display `314.1592654`)

#1 SHIFT → #9 π → #1 SHIFT → #5 ✱ → #16 1 → #21 0 → #23 x^2 → #1 SHIFT → #25 =

47. 19.63495408 (display `19.63495408`)

#1 SHIFT → #9 π → #1 SHIFT → #5 ✱ → #17 2 → #22 • → #12 5 → #23 x^2 → #1 SHIFT → #25 =

48. 153.93804003 (display `153.93804`)

#1 SHIFT → #9 π → #1 SHIFT → #5 ✱ → #6 7 → #23 x^2 → #1 SHIFT → #25 =

49. 4.52389342 (display `4.523893421`)

#1 SHIFT → #9 π → #1 SHIFT → #5 ✱ → #16 1 → #22 • → #17 2 → #23 x^2 → #1 SHIFT → #25 =

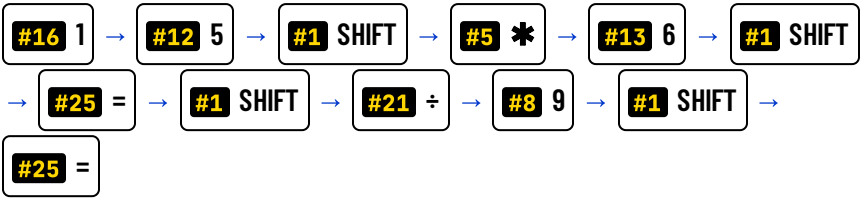
50. 120 (display `120`)

#16 1 → #21 0 → #1 SHIFT → #19 *nCr* → #18 3 → #1 SHIFT → #25 =

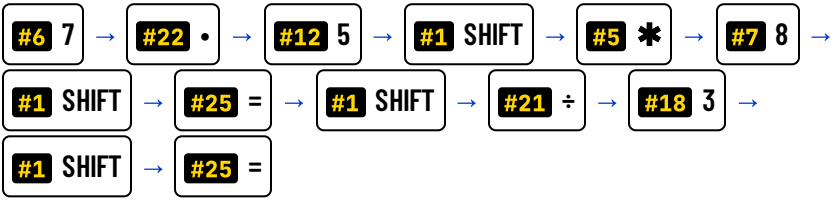
51. 1 (display `1`)

#13 6 → #1 SHIFT → #19 *nCr* → #13 6 → #1 SHIFT → #25 =

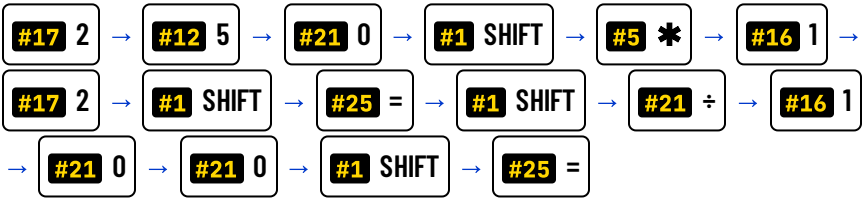
52. 10 (display `10`)



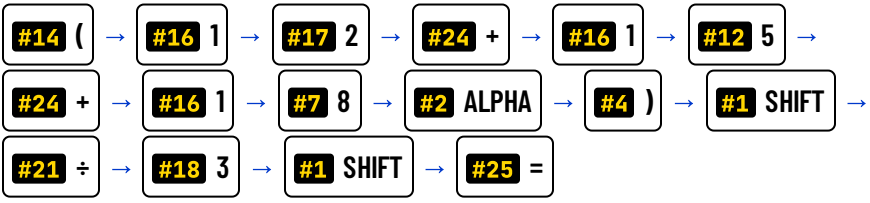
53. 20 (display `20`)



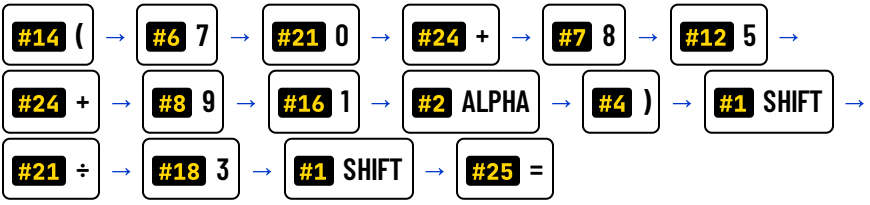
54. 30 (display `30`)



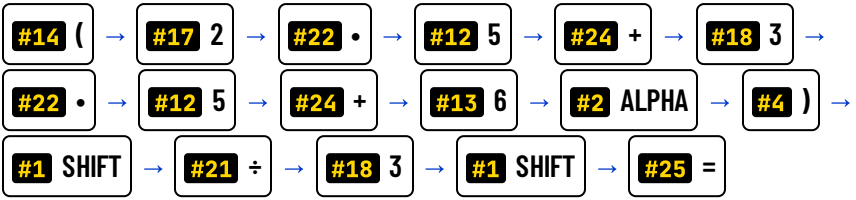
55. 15 (display `15`)



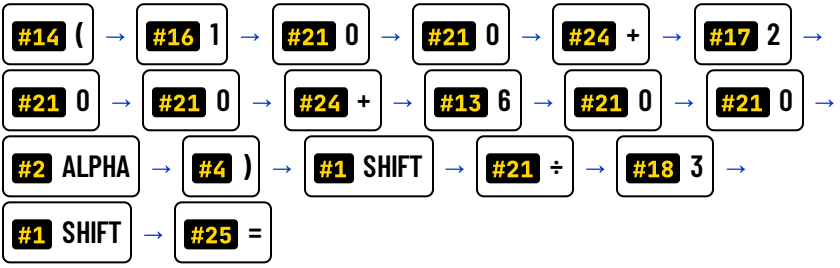
56. 82 (display `82`)



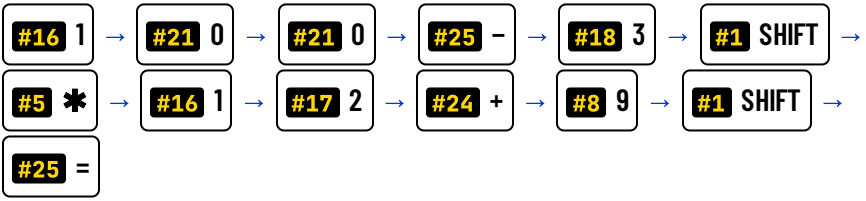
57. 4 (display `4`)



58. 300 (display `300`)



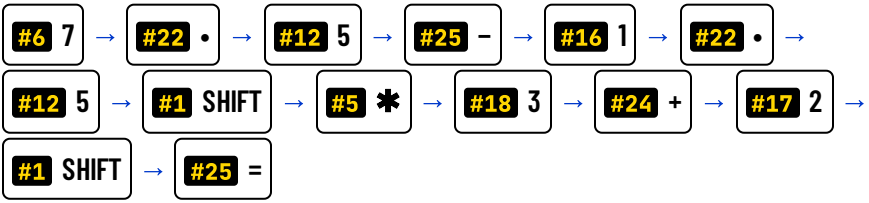
59. 73 (display `73`)



60. 48 (display `48`)



61. 5 (display `5`)



62. 42 (display 42)

#13 6 → #11 4 → #25 − → #7 8 → #1 SHIFT → #5 ✱ →
#11 4 → #24 + → #16 1 → #21 0 → #1 SHIFT → #25 =

63. 255 (display 255)

#18 3 → #21 0 → #21 0 → #25 − → #17 2 → #12 5 →
#1 SHIFT → #5 ✱ → #11 4 → #24 + → #12 5 → #12 5 →
#1 SHIFT → #25 =

64. 1 (display 1)

#8 9 → #25 − → #18 3 → #1 SHIFT → #5 ✱ → #18 3 →
#24 + → #16 1 → #1 SHIFT → #25 =

65. 35 (display 35)

#11 4 → #12 5 → #25 − → #12 5 → #1 SHIFT → #5 ✱ →
#12 5 → #24 + → #16 1 → #12 5 → #1 SHIFT → #25 =

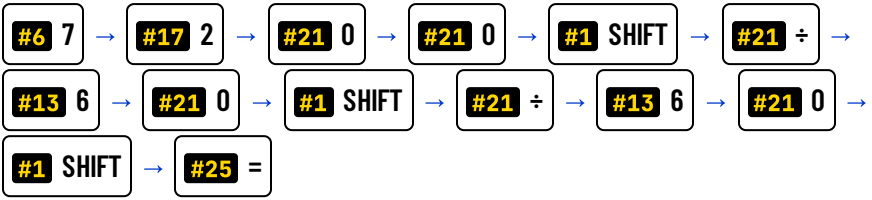
66. 25 (display 25)

#16 1 → #21 0 → #21 0 → #21 0 → #1 SHIFT → #21 ÷ →
#7 8 → #1 SHIFT → #21 ÷ → #12 5 → #1 SHIFT → #25 =

67. 16 (display 16)

#8 9 → #13 6 → #1 SHIFT → #21 ÷ → #17 2 → #1 SHIFT
→ #21 ÷ → #18 3 → #1 SHIFT → #25 =

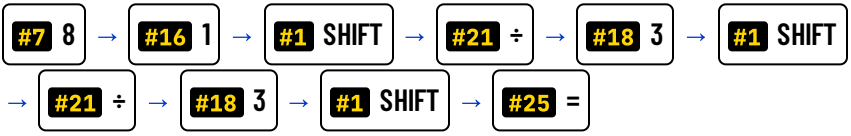
68. 2 (display `2`)



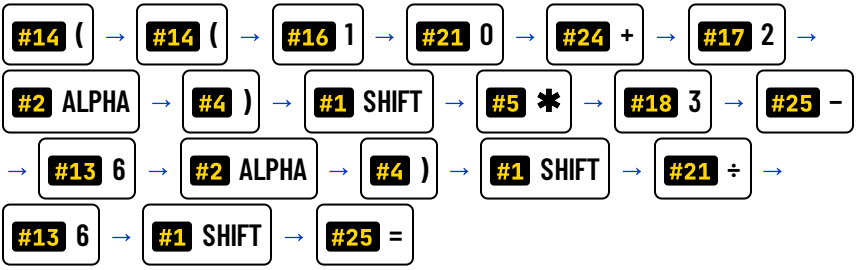
69. 5 (display `5`)



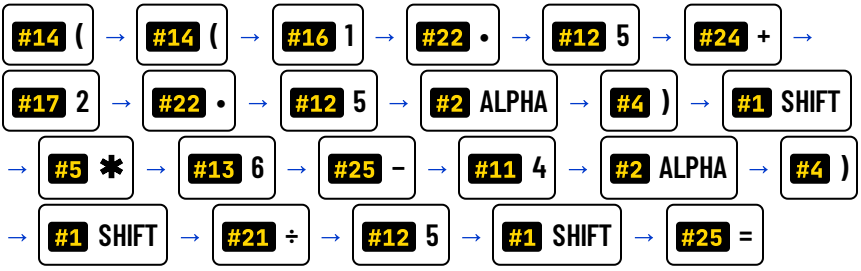
70. 9 (display `9`)



71. 5 (display `5`)



72. 4 (display `4`)



73. 25 (display 25)

#14 (→ #14 (→ #17 2 → #21 0 → #24 + → #18 3 → #21 0 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #17 2 → #25 − → #17 2 → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #18 3 → #1 SHIFT → #25 =

74. 10 (display 10)

#14 (→ #14 (→ #6 7 → #24 + → #7 8 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #8 9 → #25 − → #16 1 → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #16 1 → #17 2 → #1 SHIFT → #25 =

75. 24 (display 24)

#16 1 → #17 2 → #24 + → #18 3 → #1 SHIFT → #5 ✱ → #11 4 → #1 SHIFT → #25 =

76. 5 (display 5)

#14 (→ #16 1 → #17 2 → #24 + → #7 8 → #2 ALPHA → #4) → #1 SHIFT → #21 ÷ → #11 4 → #1 SHIFT → #25 =

77. 49 (display 49)

#14 (→ #18 3 → #24 + → #11 4 → #2 ALPHA → #4) → #1 SHIFT → #5 ✱ → #14 (→ #8 9 → #25 − → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #25 =

78. 144 (display `144`)

#16 1 → #17 2 → #23 x^2 → #1 SHIFT → #25 =

79. 12 (display `12`)

#1 SHIFT → #23 √ → #16 1 → #11 4 → #11 4 → #1 SHIFT → #25 =

80. 1 (display `1`)

#1 SHIFT → #8 tan → #11 4 → #12 5 → #2 ALPHA → #4) → #1 SHIFT → #25 =

81. 3 (display `3`)

#10 log → #16 1 → #21 0 → #21 0 → #21 0 → #2 ALPHA → #4) → #1 SHIFT → #25 =

82. 0 (display `0`)

#1 SHIFT → #14 ln → #16 1 → #2 ALPHA → #4) → #1 SHIFT → #25 =

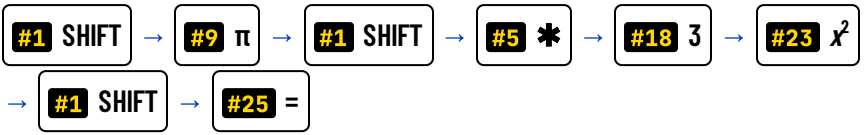
83. 100 (display `100`)

#1 SHIFT → #10 10^x → #17 2 → #2 ALPHA → #4) → #1 SHIFT → #25 =

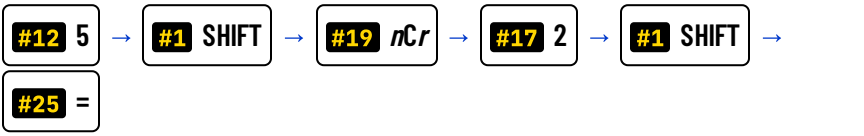
84. 1800 (display `1800`)

#13 6 → #9 ×10^x → #17 2 → #1 SHIFT → #5 ✱ → #18 3 → #1 SHIFT → #25 =

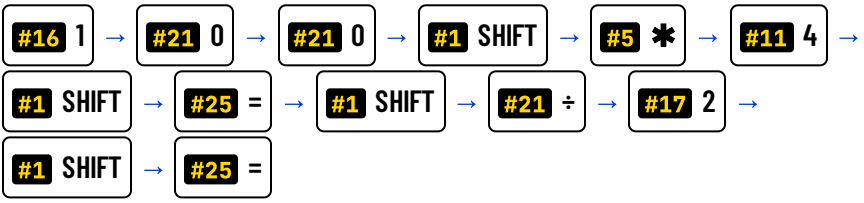
85. 28.27433388 (display `28.27433388`)



86. 10 (display `10`)



87. 200 (display `200`)



88. 9 (display `9`)



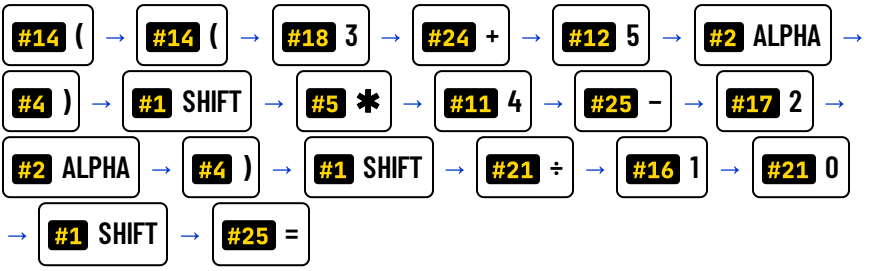
89. 33 (display `33`)



90. 10 (display `10`)



91. 3 (display 3)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #9
7. #10
8. #14
9. #15
10. #19
11. #20
12. #22
13. #23
14. #24
15. #25
16. #1 SHIFT then #3 ◀
17. #2 ALPHA then #3 ◀
18. #2 ALPHA then #4 SETUP
19. #1 SHIFT then #5 OPTN
20. #1 SHIFT then #6 7
21. #1 SHIFT then #7 8
22. #1 SHIFT then #8 9
23. #1 SHIFT then #9 ×10x
24. #1 SHIFT then #10 log
25. #1 SHIFT then #11 4
26. #1 SHIFT then #12 5
27. #1 SHIFT then #13 6
28. #1 SHIFT then #14 (
29. #1 SHIFT then #15 DEL
30. #1 SHIFT then #16 1
31. #1 SHIFT then #17 2
32. #1 SHIFT then #18 3
33. #1 SHIFT then #19 ×
34. #1 SHIFT then #20 AC/ON
35. #1 SHIFT then #21 0
36. #1 SHIFT then #22 •
37. #1 SHIFT then #23 x^2
38. #1 SHIFT then #24 +
39. #1 SHIFT then #25 -

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	×10x	#9
10^x — antilog	10x	#1 → #10
Addition +	+	#24
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#2
Ans — recall the last answer into the current expression	Ans	#1 → #24
Closing bracket). Closing brackets at the end of a line may be omitted	)	#2 → #4
Combinations nCr	nCr	#1 → #19
Common logarithm $\log_{10}$	log	#10
Cosine (current angle unit)	cos	#1 → #7
Cursor ▶ — move right in the line	▶	#1 → #3
Cursor ◀ — move left in the line	◀	#3
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	•	#22
DEL — delete the character at the cursor	DEL	#15
Digit 0	0	#21
Digit 1	1	#16
Digit 2	2	#17
Digit 3	3	#18
Digit 4	4	#11
Digit 5	5	#12
Digit 6	6	#13
Digit 7	7	#6

Digit 8	8	#7
Digit 9	9	#8
Division ÷	÷	#1 → #21
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	=	#1 → #25
Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh	hyp	#1 → #11
INS — insert mode at the cursor	INS	#1 → #15
Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#1 → #13
Inverse sine $\sin^{-1}$	$\sin^{-1}$	#1 → #12
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#1 → #22
MODE — choose calculation mode	MENU	#2 → #3
Multiplication ×	✱	#1 → #5
Multiplication ×	×	#19
Natural logarithm ln	ln	#1 → #14
OFF — power off (state is lost; press ON to restart)	OFF	#1 → #20
ON / AC — power on, or clear all	AC/ON	#20
Opening bracket (	(	#14
OPTN — options for the current mode	OPTN	#5
Polar → rectangular Rec(r,θ)	Rec(	#1 → #18
Rectangular → polar Pol(x,y)	R↔P	#1 → #16

Rectangular → polar Pol(x,y)	**Pol(**	**#1 → #17**
SETUP — angle unit, number format, display options	**SETUP**	**#4**
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	**SHIFT**	**#1**
Sine (uses the current angle unit)	**sin**	**#1 → #6**
Square root √	**√**	**#1 → #23**
Square x^2	**x^2**	**#23**
Subtraction −	**−**	**#25**
Tangent (current angle unit)	**tan**	**#1 → #8**
π (3.14159265…) as an exact constant	**π**	**#1 → #9**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

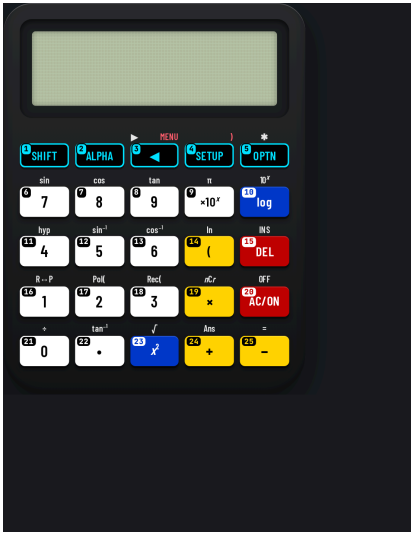
This compact model has no dedicated ÷,) or = keys in the photo, so: SHIFT 0 = ÷, ALPHA SETUP =), SHIFT − = "=" (EXE), SHIFT + = Ans, SHIFT x^2 = √. The physical keyboard Enter, /,) keys also work. Several duplicated labels were rationalised.

Planned (not yet working)

- Statistics mode
- Base-N mode
- Equation solver
- Fraction template
- Memory M+ / M−
- Table mode
- Constants
- Unit conversions
- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 SHIFT	**#2** ALPHA	**#3** ◀ ⇧ ▶
#4 SETUP	**#5** OPTN ⇧ ✱	**#6** 7 ⇧ sin
#7 8 ⇧ cos	**#8** 9 ⇧ tan	**#9** ×10x ⇧ π
#10 log ⇧ 10x	**#11** 4 ⇧ hyp	**#12** 5 ⇧ $\sin^{-1}$
#13 6 ⇧ $\cos^{-1}$	**#14** (⇧ ln	**#15** DEL ⇧ INS
#16 1 ⇧ R↔P	**#17** 2 ⇧ Pol(	**#18** 3 ⇧ Rec(
#19 × ⇧ nCr	**#20** AC/ON ⇧ OFF	**#21** 0 ⇧ ÷
#22 • ⇧ $\tan^{-1}$	**#23** x^2 ⇧ √	**#24** + ⇧ Ans
#25 − ⇧ =		

U-45 CALCULATORS · Calculator #3 · cut out and keep